

Project Violet Milk

Prototype explainer and presentation script

Tracing cyber-fraud proceeds from an FIR to a court dossier and a production order

Case demonstrated	CP-CYBER-2026-001
Reported loss	INR 4,70,000
Chains traced	Ethereum, Tron (TRC-20)
Automated tests	224 passing
API surface	23 paths
Licence	MIT - free for any force to use
Cost per district	Nil. No licence, no cloud account, no subscription.
Prepared	07 September 2026

This document has two halves. The first is the spoken presentation, written to be delivered standing, in about six minutes. The second explains every part of the system in the order a technical reviewer would want to inspect it.

Part One - the presentation

Approximately six minutes, delivered standing

Deliver the opening without touching the screen. The single most persuasive thing in this presentation is not a feature; it is that the system distinguishes what it knows from what it has guessed, and says so out loud before anyone asks.

OPENING - a specific human being

Sir, a resident of Chandigarh lost four lakh seventy thousand rupees to a digital arrest fraud. She did everything correctly. She called 1930 within the hour.

By then it was already too late, and I want to show you exactly how much too late. Her money left her bank account at ten thirty-one and four seconds. At ten thirty-two and two seconds - **fifty-eight seconds later** - it was USDT on a public blockchain, outside Indian banking, and outside anything the helpline can see.

THE GAP - name it precisely

The Citizen Financial Cyber Fraud Reporting and Management System can freeze her bank account, and it is good at that. But at that fifty-eighth second the trail goes dark, and the officer who must write the chargesheet has nothing to work with.

Not because he lacks the law. Section 94 of the Bharatiya Nagarik Suraksha Sanhita lets him compel any exchange in the country to produce its records. He lacks one thing only: he does not know *which* exchange, or *which* deposit address to name in that order.

That is the entire gap. It is not a legal gap and it is not a policy gap. It is a capability gap, and it is narrow enough to close.

WHAT WE BUILT - one sentence

So we built the thing that begins where 1930 stops, and ends with a signed piece of paper.

CREDIBILITY - lead with the dashed line

One thing before I show you anything on the screen. On our map, every solid line is a bank record or a blockchain transaction hash that you can paste into a public explorer and check yourself. One line is dashed.

That one we *inferred* - a thirteen-second gap between a UPI debit and an exchange deposit. It is a hypothesis, not a proven transfer. It is drawn differently on the screen, typed differently in the data, and printed as INFERRED on the face of the court document. Sixteen confirmed, one inferred.

I mention it first because everything else I say is only worth something if you believe that distinction is real.

THE MAP - show, do not narrate

This is the money. Victim, mule bank account, exchange, and then the wallet that received the USDT. That wallet scores sixty-five out of a hundred, and I can tell you exactly why: twenty points because it is the case seed, fifteen because funds left three minutes after they arrived, fifteen for a direct transfer into a mixer, and fifteen because a cross-chain bridge sits two hops downstream.

Seven rules, fixed weights, evidence cited against each one. No model produced that number. A defence lawyer can check the addition, and I can explain it in a witness box.

THE TURN - show the honest negative

Now I will show you something no other prototype today will show you. Our own tool will tell you it has *failed*.

Sixty per cent of the tainted funds land in an exchange pool that already held clean money. The traced share falls to twelve per cent. Our own threshold is thirty per cent, so the system says NOT FLAGGED. That pool contains real criminal proceeds and we still report that the trail ends there.

A tool that only ever says guilty is worthless in a courtroom. That negative result is what makes the sixty-five on the previous screen worth believing.

THE AI QUESTION - answer before it is asked

You will want to know where the artificial intelligence is. Here it is, and look what it did. It flagged the *complainant*, because she moved the largest single amount and that makes her a statistical outlier.

Her risk score is zero. The machine learning produces investigative leads on a separate screen with a separate shape, and it cannot touch a score. That is enforced by an automated test, not by a promise on a slide.

CLOSING - Tuesday morning

Which brings me to the only question that actually matters. On Tuesday morning, what does the investigating officer do?

He presses one button. Out comes a draft production order under Section 94, addressed to the exchange, with the deposit address, the amount, the timestamp and the transaction hash already written into the Schedule - and every line marked CONFIRMED or INFERRED so he knows in advance what a defence will contest. He completes his designation, has it checked, signs it, and serves it.

It runs on one laptop. It works with the wifi unplugged. It costs nothing per district, and it is MIT licensed, so any force in this country can take it and use it without asking us.

We are not asking you to believe it is finished. We are asking for three months in one cyber cell and one number back: how many cases where the trail continued past the bank. If that number is not meaningfully above zero, we have not earned anything.

Part Two - every aspect of the system

In the order a reviewer would inspect it

1. The case being demonstrated

A single synthetic case, built to encode a real laundering pattern rather than to flatter the software. Every figure below is reproduced exactly by the engines on every run, and an automated test fails the build if any of them drift.

Time (IST)	Event	Basis
10:31:04	Complainant to mule account, INR 4,70,000, UTR 420192830192	CONFIRMED
10:31:17	Mule account to exchange deposit address, 13 seconds later	INFERRED
10:32:02	Exchange to seed wallet, 5,200 USDT at the locked rate of INR 90.38	CONFIRMED
10:35-10:36	Seed disperses to exactly five wallets, one directly into a mixer	CONFIRMED
10:38-10:39	Dispersal Wallet 1 splits into six smaller amounts - structuring	CONFIRMED
10:41:12	2,000 USDT into a pool already holding 8,000 clean	CONFIRMED
10:44:19	Final hop into the mixer. Mixer total: 1,900 USDT	CONFIRMED

Fourteen entities, seventeen transfers, thirteen minutes and fifteen seconds from first debit to final hop.

2. Architecture

A **DataSource** interface with three implementations. The engines never learn which one is active, which is why a judge can hand over a live address mid-demonstration and nothing downstream changes.

Layer	Components
Sources	SyntheticSource (bundled case), EtherscanSource (Ethereum), TronSource (TRC-20)
Engines	graph, risk, dilution, timeline, anomaly, assets, report, STR, notice, pipeline
Services	audit, auth, anchor, ingestion, column mapper, narrative, LLM client, hashing, formatting
Interface	React single-page application, served by the same process on the same port

An address routes itself: a value beginning 0x is Ethereum, one beginning T is Tron. One command, one port, one SQLite file. No container, no database server, no cloud account.

3. The eleven screens, and what each is for

Screen	What it answers
Command Center	Opens on a worked case. Highest-risk entities and the recent custody trail.
Case Intake	FIR and NCRP references, complainant loss, seed address.
Evidence Ingestion	Upload a bank statement or transfer list. Rows merge into the case graph as typed entities.
Graph Visualiser	The map. Solid is proven, dashed is inferred. Click any entity for a card an officer can act on.
Asset Ledger	Which currency moved, at which layer of the trace, and what it is in rupees.
Timeline	The same events as a clock. This is what makes the fifty-eight seconds land.
Risk Inspector	Opens on the whole case, then the point-by-point arithmetic for any one entity.
Dilution Calculator	How much of what sits at an address is traceable to the complainant.
Flag Agent	Unsupervised model output, kept deliberately apart from scoring.
Sec 63 BSA Dossier	The court document, the STR draft, and the production order draft.
Chain of Custody	Every action, hash-chained, with a verify button that names any broken link.

4. The seven rules

Section 63 of the Bharatiya Sakshya Adhiniyam 2023 requires explaining how an electronic record was produced. Arithmetic can be explained to a court; a neural network cannot be cross-examined. That is the whole reason scoring is deterministic.

Rule	Points	Fires when
R1	+20	Within one hop of the case seed
R2	+15	Funds leave within 240 seconds of arriving
R3	+15	Interaction with a known mixer contract
R4	+15	A cross-chain bridge within two hops downstream
R5	+10	Fan-out to more than five addresses
R6	+10	Structuring - repeated amounts under a reporting band
R7	+15	Time correlation with the victim debit. Suppressed when R1 fires, to avoid counting proximity twice.

Bands: 0-24 LOW, 25-49 MEDIUM, 50-74 HIGH, 75-100 CRITICAL. The seed scores 65. An automated test asserts that every score equals the clamped sum of its own cited indicators, so a figure that did not come from these rules would fail the build.

Honest limitation
The weights are reasoned, not calibrated. There is no empirical basis for twenty points rather than fifteen, and no labelled ground truth exists to measure a false positive rate against. What can be defended is that every score is reproducible and contestable: an accused can point at rule R4 and argue the bridge inference, which is impossible against an opaque model.

5. Dilution - and why there are two models

When tainted funds enter an address that already holds clean money, the traceable share falls. Proportional haircut is used because it makes the fewest assumptions a defence can attack: it says only that a pool mixes, whereas first-in-first-out requires assumptions about ordering.

Hop	Prior clean	Tainted in	Traceable share
Victim account	-	INR 4,70,000	100%
Seed wallet	0	5,200 USDT	100%
Layering wallet	1,200	1,800	60%
Exchange pool	8,000	2,000 at 60%	12% - NOT FLAGGED

The second model exists because the first cannot honestly be used on a live chain. A public explorer cannot tell you an address's balance at the moment funds arrived, so there is no denominator. Rather than print a confidently wrong ratio, live traces switch to a propagation model that carries a caveat stating in plain words that it measures reach, not proportion.

6. Evidence, custody and anchoring

Each custody entry hashes the entry before it, so altering any row breaks every hash that follows. The verify endpoint names the first broken link. A document's own digest is recorded separately, because a file cannot contain its own hash - that detached record is what makes a dossier verifiable afterwards.

```
entry_hash = SHA256(prev_hash | id | time | user | action | target)
```

What this proves	What it does not
That the log has not been altered without detection, and that a specific document existed in a specific state at a specific time.	That the document is true. Anyone with write access to the database could recompute the chain forward from an edit, which is why the wording everywhere is <i>detectable</i> , never tamper-proof.

Optional on-chain anchoring closes that gap by placing the digest somewhere outside our control. It proves existence and time. It does not prove authorship or truth, and the dossier says so in those words.

7. Identity - where a hex string becomes a person

This is the step most crypto-tracing demonstrations skip. A wallet address is not a person and this system never claims otherwise; the STR draft prints NOT IDENTIFIED in those words.

The lawful route runs through the exchange holding the KYC record, and the instrument is a written order under Section 94 BNSS 2023. One button assembles that order's recitals from the trace: the addressee, the deposit address, the amounts, the timestamps and the transaction hashes, together with the specific records sought - KYC, linked Indian bank accounts and UPI handles, IP and device logs, and the present status of any balance.

Safeguard	How it is enforced
Never a signed order	Status is permanently DRAFT. A test walks the API schema and fails the build if any route appears that could issue or serve one.
Never invents an addressee	The legal name and registered address are placeholders. We know an on-chain label, not a company's identity, and serving the wrong entity is a real harm.
Names nobody when nothing was found	A trace that reached no exchange says so, rather than nominating the nearest address.
Ranks by tainted value	A large clean inflow cannot make an address the right one to serve.
Declares its evidence	Every line of the Schedule is marked CONFIRMED or INFERRED on the face of the order.

8. Where the artificial intelligence is, and is not

Component	Role	Touches a score?
Language model	Drafts the officer's narrative paragraph and guesses bank CSV column headings	No
Isolation Forest	Flags behavioural outliers as investigative leads	No
DBSCAN	Clusters wallets with similar behaviour	No
Seven rules	Produce every risk figure in the system	Yes - only these

The proof, rather than the promise. The anomaly model flags the complainant, because she moved the largest single amount and is therefore a statistical outlier. Her risk score is zero. If the model influenced scoring, the victim would be the prime suspect. That automated test is the reason the model is kept off the score.

Every AI feature degrades to a deterministic path. One test deletes the API key *and* the cache, then generates a complete dossier from the template and the heuristic column mapper. If the provider is unreachable on the day, nothing stops.

9. Live mainnet, and surviving a dead network

- Any Ethereum or Tron address can be pasted in and traced against the real chain, using the same rules, the same custody log and the same documents.
- Calls are self-throttled to three per second and every response is written to disk.
- A circuit breaker trips after three consecutive failures, so a dead network falls through to cache in about a second rather than sixty.
- A stage mode replays pre-tested addresses from disk without any network call at all. The interface still reports the source as cached, so a replay can never be presented as a live query.
- Traces are bounded - depth three, capped edges per entity, with each asset given a fair share of that budget so large native-currency figures cannot starve token transfers out. When a bound is reached the interface reports it rather than

implying completeness.

10. Cost and practical feasibility in India today

Component	What is used	Cost
Chain data	Etherscan free tier	Nil
Tron data	TronGrid free tier	Nil
Database	SQLite - a single file	Nil
Software	Python, FastAPI, React - all open source	Nil
AI narrative	Optional; template fallback is tested	Nil
Hardware	One laptop already on the desk	Existing

The marginal cost of adding another district is nil. That is the argument - not that this is cheaper than the commercial tools, but that it sits in an entirely different category of purchase decision. There is no per-seat licence to renew and no procurement cycle to survive.

11. Where this sits among systems that already exist

System	What it does	Where it stops
NCRP / 1930	Complaint intake and routing	Not an analysis tool
CFCFRMS	Golden-hour bank-to-bank freeze on UTRs	The moment funds become crypto
CCTNS	FIR and case records	No financial tracing
Commercial tracing	Address attribution at scale	Per-seat foreign-currency licence; cloud dependent; no Indian statutory output
This system	FIR to trace to Sec 63 dossier to Sec 94 production order, offline	Attribution - fifteen curated labels, stated in writing

We do not compete with commercial attribution and should not pretend to. Their product is address identification at scale, priced accordingly, which is why it sits at national agencies rather than in a district cyber cell. If a force ever buys that data, this workflow consumes it. An unlabelled address here renders as *unknown*, never as clean.

12. What this system does not claim

Every item below is printed in the repository, in the generated documents, or both. Volunteering them is cheaper than being caught by a question.

- 1. The rule weights are not calibrated.** They are explainable, which is the point; they are not validated against outcomes.
- 2. The false positive rate is unknown.** There is no labelled ground truth, so any figure would be invented.
- 3. Bank clocks and block clocks are not the same clock.** Time correlation across the two is a lead, and is drawn as a dashed edge for exactly that reason.
- 4. Bitcoin is not supported.** The data model is account-based; Bitcoin is UTXO and needs a different graph plus a change-address heuristic that is itself a source of error.
- 5. Attribution is fifteen curated labels.** Absence of a label means unknown, never clean.
- 6. Chain reorganisations are not handled.** A trace records what the explorer returned at that moment, which is why the custody log timestamps it.
- 7. The statutory wording is unreviewed.** The Section 63 certificate and Section 94 order were drafted in good faith by people who are not lawyers, and say so on their face.
- 8. The interface is English only.** A bilingual interface is a fortnight of work and has not been done.
- 9. It is not affiliated with Chandigarh Police.** Stated in the first lines of the repository and in a permanent banner on every screen.

13. What making this official would require

- 1. Legal review of the statutory wording,** which everything downstream depends on.
- 2. A CERT-In empanelled security audit,** the standard prerequisite for handling case data on government infrastructure.
- 3. A decision under the Digital Personal Data Protection Act 2023** on the AI narrative - either an in-house model, or shipping with the tested template path and no external transmission at all.
- 4. Integration with CCTNS and NCRP,** so an officer reads the FIR rather than retyping it.

5. An attribution feed - realistically an arrangement with FIU-IND registered Indian exchanges, or a commercial licence layered underneath this workflow.
6. Per-officer accounts and keys held in a hardware security module. Today there is one seeded officer and a default password the application itself warns about.
7. A three-month pilot in one cyber cell, measured on one number: cases where the trail continued past the bank boundary.

Closing note

This is an independent student prototype built for a hackathon. It is not affiliated with, endorsed by, or deployed by Chandigarh Police, FIU-IND or any law-enforcement agency. Agency names appear only to describe the intended user of a prototype. All demonstration identities, accounts, UPI handles and addresses are synthetic. The software produces investigative leads and supporting paperwork; it does not constitute a finding of guilt and does not identify any person without independent verification under Section 94 BNSS 2023.